

Video Transcripts

Week 5.1: Gen AI and Disruptive Innovation: Gen AI in Marketing and Operations

Video 1: What Makes an Innovation Disruptive?

Hello learners, and welcome to this week. A useful way to think about disruption is this: disruptive innovations rarely begin by being better – they begin by changing who can participate in a market. They alter the rules of competition, often by making capabilities cheaper, more accessible, or easier to use.

Take the transformation of the media industry. For decades, television networks competed by investing heavily in content production, broadcast infrastructure, and prime-time programming. Distribution power determined success. When Netflix shifted from DVD rentals to streaming in 2007, it did not initially outperform broadcasters on production budgets. What it changed was the delivery model. Instead of fixed schedules, viewers could watch content on demand. Over time, convenience, global distribution, and algorithmic recommendations reshaped how audiences consumed entertainment. Today, streaming has fundamentally restructured the television industry.

A similar pattern appeared in hospitality. Global hotel chains spent decades improving service quality, brand consistency, and loyalty programmes. Then Airbnb introduced a platform that allowed ordinary homeowners to rent spare rooms or apartments. The company did not build hotels or operate traditional hospitality infrastructure. Instead, it unlocked millions of underutilised properties and turned them into accommodation supply. Within a decade, Airbnb had listings in more cities than the largest hotel groups. The disruption came not from better hotels, but from expanding who could provide lodging. This same dynamic is emerging with generative AI.

Many organisations currently use AI tools to improve productivity – developers write code faster, marketers generate campaign drafts, and analysts summarise reports quickly. These are valuable efficiency gains. But the deeper disruption occurs when expert capabilities become embedded in software.

A startup with a small team can now design products, generate marketing assets, build prototypes, and analyse markets using AI systems that previously required specialised departments.

When expertise becomes widely accessible, barriers to entry fall and new competitors emerge. That is the essence of disruptive innovation: not just better technology, but a shift in who can compete in the market.

Video 2: Generative AI as a General-Purpose Technology

Some technologies do not stay confined to one product or one department. Instead, they spread across industries and reshape how organisations operate. Economists call these general-purpose

technologies – innovations like electricity, the internet, and cloud computing that become foundational infrastructure for the economy.

Generative AI is beginning to follow a similar trajectory. A useful way to understand this is by looking at how enterprise software platforms are integrating AI directly into everyday business workflows.

Consider Salesforce, one of the world's largest providers of customer relationship management software. Its platform is used by thousands of companies to manage sales pipelines, customer service interactions, and marketing campaigns. Instead of offering generative AI as a separate tool, Salesforce embedded it directly into the platform through its AI layer known as Einstein GPT.

This allows sales teams to generate personalised outreach emails automatically based on customer history stored in the CRM system. Customer service teams can instantly summarise long support conversations and generate draft responses. Marketing teams can create campaign content using customer data already stored in the platform.

In other words, generative AI is not operating as a standalone application. It becomes an intelligence layer integrated into existing enterprise systems.

This has major implications for how organisations operate. When intelligence is embedded inside widely used platforms, the impact spreads quickly across departments. A single technology begins influencing sales, customer service, marketing, and analytics simultaneously.

We see similar patterns across other platforms as well. E-commerce provider Shopify now offers AI tools that help merchants generate product descriptions and marketing copy automatically. Small businesses can produce professional-quality content without dedicated marketing teams.

What connects these examples is diffusion. Generative AI is not confined to a single industry or task. It is spreading across enterprise software, digital commerce, consulting, media, and research.

That pattern is exactly what defines a general-purpose technology. Its value does not come from one breakthrough application. It comes from being embedded everywhere, gradually transforming how organisations create, deliver, and scale value.

Video 3: From Product-Centric to Intelligence-Centric Firms

In the past, competitive advantage came from building better products. Today, it increasingly comes from building systems that continuously learn from users. In other words, firms are moving from shipping finished products to operating intelligence platforms that evolve in real time.

Generative AI is beginning to shift that model. Increasingly, companies are moving from selling static products to building intelligence systems that continuously adapt to users. A clear example of this transition can be seen at Duolingo.

For many years, Duolingo operated like a traditional digital product. The language-learning app offered structured lessons, exercises, and quizzes designed by human curriculum teams. Every learner essentially progressed through the same predefined content pathway.

Generative AI is transforming that model. In 2023, Duolingo introduced Duolingo Max, a premium tier powered by models from OpenAI. Instead of simply completing exercises, learners can now interact with AI-driven features such as Roleplay, where users hold simulated conversations with AI characters in the language they are learning.

For example, a learner studying Spanish might practise ordering food at a restaurant or asking for directions in a city. The AI generates responses dynamically, adjusts the conversation based on the learner's input, and provides feedback explaining mistakes or suggesting alternative phrasing.

This means the product is no longer just a fixed set of lessons. The learning experience is generated in real time, adapting to the learner's skill level, mistakes, and conversational style. The shift has major implications for the company's operating model. Instead of scaling learning through more static content, Duolingo scales through intelligence infrastructure capable of generating millions of personalised learning interactions every day.

This illustrates the broader transition from product-centric firms to intelligence-centric firms. In product-centric models, value is embedded in the offering itself. In intelligence-centric models, value emerges from systems that continuously generate insights, adapt experiences, and improve through data.

In the AI era, the most successful organisations may not simply build better products. They will build products that learn, evolve, and generate new value through embedded intelligence.

Video 4: Case Study: Bloomberg

One of the defining effects of disruptive technologies is that they blur the boundaries between industries. Companies that once operated in clearly defined sectors begin expanding into adjacent roles. Generative AI is accelerating this process by making intelligence – analysis, synthesis, and insight generation – programmable and scalable. A useful way to understand this shift is through the evolution of Bloomberg.

For decades, Bloomberg was known primarily as a financial information provider. Its core product, the Bloomberg Terminal, delivered market data, news, and analytics to traders, analysts, and financial institutions. In this model, industry roles were relatively clear. Media companies produced news, data providers delivered information feeds, and financial analysts interpreted that information to generate insights. However, generative AI is beginning to collapse these distinctions.

In 2023, Bloomberg introduced BloombergGPT, a large language model trained on decades of financial documents, market data, news articles, and filings. The model contains around 50 billion parameters and is designed specifically for financial tasks such as analysing earnings reports, summarising regulatory filings, classifying financial news, and generating market insights. What makes this

significant is not just the technology itself, but how it reshapes Bloomberg's role in the financial ecosystem.

Traditionally, Bloomberg provided information and analytical tools that professionals used to conduct their own analysis. With generative AI, the platform increasingly performs parts of that analysis automatically. Traders can query complex datasets through natural language, analysts can generate summaries of earnings calls within seconds, and financial research workflows become significantly faster. In effect, Bloomberg is no longer only a data provider or media organisation. It is becoming an AI-powered financial intelligence platform that combines information, analytics, and automated insight generation within a single interface.

This illustrates how generative AI collapses industry boundaries. A company that once sat at the intersection of media and financial data now moves closer to roles traditionally associated with research analysts, advisory firms, and financial technology platforms. For leaders, this shift carries an important implication. Firms will increasingly compete not only with traditional industry peers, but with organisations from adjacent sectors that can combine data, software, and AI-driven intelligence into unified platforms.

In the AI era, industries do not simply evolve. Their boundaries begin to dissolve.

Video 5: Zero-Marginal-Cost Knowledge Businesses

One of the most disruptive economic implications of generative AI lies in how it reshapes the cost structure of knowledge work. Traditionally, knowledge-intensive industries such as consulting, research, law, and financial advisory were constrained by human capacity. If a firm wanted to serve more clients, produce more analysis, or answer more questions, it had to hire more experts. Output scaled with headcount. As a result, marginal costs remained high – each additional report, recommendation, or client interaction required incremental professional effort.

Generative AI is beginning to change that model. A well-documented example comes from Morgan Stanley, one of the world's largest wealth management firms, with over 16,000 financial advisors serving millions of clients. Advisors constantly rely on internal research – market outlook reports, portfolio guidance, regulatory updates, and investment analyses produced by Morgan Stanley's research teams. Historically, locating the right information required manually searching through large document repositories or contacting internal specialists.

In 2023, Morgan Stanley introduced an AI-powered knowledge assistant built on technology from OpenAI. The system was trained on more than 100,000 internal research documents and knowledge resources used across the firm's wealth management division. Advisors can now ask complex questions conversationally. For example:

'What is the firm's latest view on US interest rates?'

'How should we advise clients concerned about recession risk?'

'What guidance does our research provide on energy sector exposure?'

The system retrieves relevant documents, synthesises insights, and generates structured responses grounded in Morgan Stanley's proprietary research. The economic shift becomes clear when we consider scale. Once the AI assistant is deployed, thousands of advisors can access the same knowledge simultaneously. The firm does not need to expand research teams proportionally as client interactions increase. The incremental cost of generating one additional answer or explanation becomes extremely small.

This is what economists describe as near-zero marginal cost knowledge production. However, this does not eliminate human expertise. Instead, it redistributes it. Research teams still produce original insights, but generative AI dramatically amplifies the distribution of that expertise across the organisation.

The strategic result is powerful. Instead of scaling advisory capacity purely through hiring, firms can scale through intelligence infrastructure that replicates institutional knowledge instantly. In the emerging AI economy, competitive advantage may increasingly belong to organisations that transform expert knowledge into scalable digital systems rather than purely human services.

Video 6: Hyper-Personalized Products

For many years, personalisation in business primarily meant targeted marketing. Companies segmented customers and delivered customised emails, recommendations, or advertisements. However, the underlying product remained largely the same for everyone. Generative AI is beginning to change that model. Personalisation is no longer limited to communication. It is increasingly becoming embedded within the product experience itself.

A strong example can be seen at Spotify. Spotify operates a platform with over 100 million songs and hundreds of millions of users worldwide. Helping users navigate such a vast catalogue requires more than simple recommendations. The company has long relied on machine learning to analyse listening behaviour, playlist activity, and user preferences. However, the introduction of generative AI is pushing this personalisation much further.

In 2023, Spotify launched AI DJ, a feature that uses generative AI to create a fully personalised listening session. The system analyses a user's listening history, favourite genres, and recently played tracks. It then dynamically generates a sequence of songs while also producing AI-generated voice commentary that introduces tracks, explains recommendations, and contextualises music selections. Behind the scenes, large language models and generative systems synthesise user data with Spotify's content library to generate unique listening narratives for each user.

The company has also introduced AI Playlist, where users can generate personalised playlists simply by typing prompts such as 'indie music for a rainy evening' or 'high-energy workout tracks similar to this artist'. Generative AI interprets the prompt, combines it with the user's listening patterns, and constructs a playlist that did not previously exist.

This illustrates a deeper shift. Instead of presenting the same product catalogue to every user, Spotify increasingly generates individualised product experiences in real time. The platform becomes a

system that continuously creates personalised music journeys rather than simply delivering static content.

The economic implications are powerful. Once the generative AI infrastructure is built, millions of personalised listening experiences can be generated simultaneously at minimal incremental cost. In the AI era, companies will increasingly move beyond segmenting customers to algorithmically generating unique products and experiences for each individual.

Video 7: Conversational Business Models

For most of the internet era, companies designed their digital presence around websites and apps. Customers navigated menus, browsed product categories, compared options, and clicked through pages to complete transactions. This structure assumed that users already knew where to look and how to search. Generative AI is beginning to transform this model. Instead of navigating interfaces, customers increasingly interact with companies through conversation.

A clear example of this shift can be seen at Walmart. As one of the world's largest retailers, Walmart manages millions of products across groceries, electronics, household goods, and everyday essentials. Traditionally, customers using the Walmart app had to browse categories, apply filters, and manually build their shopping cart.

In 2024, the company introduced a generative AI shopping assistant designed to simplify this process. Instead of navigating menus, customers can now describe what they want in natural language. For example, a user might ask:

'Plan a taco dinner for six people,' or
'I need a healthy lunch for the week under \$50.'

The AI system interprets the request, generates a suggested meal plan, recommends ingredients available in nearby stores, and automatically builds a shopping cart. Customers can modify the list through further conversation – swapping products, adjusting quantities, or asking for cheaper alternatives. Behind the scenes, large language models analyse the request, connect it to Walmart's product catalogue, and generate structured recommendations in real time.

The significance of this shift goes beyond convenience. It fundamentally changes the interface between the company and the customer. Instead of forcing users to understand the structure of a website, the business adapts to the intent expressed in conversation. The interaction becomes faster, more intuitive, and more personalised.

This illustrates the emergence of conversational business models. In traditional digital commerce, value is delivered through navigation and search. In conversational models, value is delivered through dialogue. AI systems interpret intent, guide decision-making, and even complete transactions within the conversation itself.

As generative AI continues to evolve, many organisations may shift from building digital platforms that customers browse to building intelligent systems that customers simply talk to.

Video 8: Dynamic Pricing via Narrative Intelligence

Pricing has traditionally been a numerical exercise. Firms estimate costs, forecast demand, and publish a price. Customers then decide whether the price feels reasonable, often without understanding the logic behind it. In many industries today, the price itself is already dynamic. Airlines, hotels, and ride-hailing platforms adjust prices continuously based on demand signals, inventory availability, and behavioural data. What generative AI introduces is a new layer: the ability to generate contextual explanations around the price in real time.

Consider the global hospitality industry. Companies such as Marriott International operate sophisticated revenue management systems that adjust room rates dynamically based on demand forecasts, seasonality, booking patterns, and remaining inventory. Historically, customers simply saw different prices on different days without understanding why.

With generative AI integrated into booking interfaces, the system can now explain those price differences in context. A business traveller searching for a weekday stay may see messaging highlighting proximity to business districts, flexible cancellation, and faster check-in. A family planning a weekend trip may instead see narrative explanations emphasising room size, bundled breakfast offers, or nearby attractions. The underlying price is still determined algorithmically through demand forecasting models. However, generative AI constructs a narrative that connects the price to perceived value for that specific customer.

Airlines operate in a similar way. Global carriers such as Emirates already adjust fares dynamically based on seat availability, booking timing, and route demand. Generative AI enables these differences to be explained more clearly. For instance, a slightly higher fare might be accompanied by contextual messaging explaining refund flexibility, improved connection reliability, or higher loyalty point accrual. Instead of presenting a number alone, the system communicates the reasoning behind the price.

This is where narrative intelligence becomes powerful. Traditional dynamic pricing optimises numbers. Narrative intelligence optimises perception and understanding. Generative AI translates complex pricing logic into explanations that align with customer priorities, improving transparency and price acceptance.

For organisations, this opens a new strategic capability. Pricing becomes not only a financial decision but also a communication system that continuously explains value. In the era of generative AI, the number is no longer the entire price. The story behind the price increasingly shapes how customers interpret and accept it.

Video 9: Digital Twins with Generative Simulation

Traditionally, organisations improved operations by analysing past data and making incremental adjustments. Leaders reviewed historical performance, ran limited forecasts, and then implemented changes in the real system. This approach often meant learning through costly trial and error.

Digital twins are changing that model.

A digital twin is a virtual representation of a physical system – such as a factory, supply chain, aircraft engine, or energy grid. When combined with generative AI, the twin becomes more than a mirror of reality. It becomes a simulation environment for scenario modelling and strategic experimentation.

Consider manufacturing operations at Siemens. In a Siemens smart factory, production equipment – robots, conveyors, assembly stations, and sensors – continuously stream operational data into a digital twin built on industrial IoT platforms. This twin replicates the behaviour of the real production line using simulation models that represent machine performance, production throughput, energy consumption, and process timing.

Generative AI enhances this system in two key ways.

The first is scenario modelling. Engineers define constraints such as production targets, machine capacity, maintenance schedules, and energy limits. Generative algorithms then simulate thousands of possible production configurations by altering machine sequencing, workflow routing, or scheduling patterns. The objective is to identify combinations that maximise output while reducing downtime and operational inefficiencies.

The second is strategic experimentation. Instead of testing operational changes directly on the factory floor, engineers experiment within the digital twin itself. For example, Siemens can simulate introducing a new product variant into an assembly line and observe how the change affects machine utilisation, cycle times, and material flow across the production system before making any physical adjustments.

Behind the scenes, these simulations combine industrial IoT data, physics-based modelling, and AI-driven optimisation algorithms. Machine learning models analyse historical performance patterns, while generative optimisation engines propose alternative operating configurations based on the defined operational goals.

The result is a continuous experimentation environment. Engineers can test hundreds of operational scenarios – new layouts, different staffing levels, alternative production schedules, or maintenance strategies – before implementing the most effective option in the real factory.

This capability fundamentally changes decision-making. Instead of relying on forecasts alone, organisations use AI-powered digital twins as strategic laboratories where operational strategies can be tested, refined, and validated before deployment.

In the era of generative AI, complex systems are no longer managed purely through experience. They are increasingly simulated, stress-tested, and optimised through intelligent experimentation.

Video 10: Autonomous Content Supply Chains

Traditionally, content production in media, publishing, and education followed a sequential pipeline – research, drafting, editing, design, and distribution. Each stage required human intervention, making

the process slow and difficult to scale. Generative AI is beginning to transform this model by enabling autonomous content supply chains, where AI systems generate, adapt, and distribute content continuously.

Let us talk about The Associated Press. The organisation has long used automated journalism systems to generate earnings reports and sports summaries from structured datasets. Financial results, game statistics, and other numerical data are ingested into natural language generation pipelines that convert structured data into readable narratives.

With modern generative AI, this process has become significantly more advanced. Instead of rule-based templates, large language models (LLMs) can interpret structured datasets and produce multiple variations of a story. For example, the same earnings dataset can generate a full article, a short summary, social media headlines, and mobile notifications simultaneously. The system can also adjust tone and complexity depending on the target audience.

Technically, these autonomous pipelines combine several components. Data ingestion systems collect structured inputs such as financial reports, sports statistics, or market data. These inputs feed into content generation models, typically LLMs fine-tuned on editorial style guidelines. Workflow automation tools then trigger downstream processes such as editing checks, metadata tagging, image generation, and distribution across websites, newsletters, and social platforms.

The education sector is adopting similar architectures. Platforms such as Udemy are integrating generative AI into learning content pipelines. AI models can generate lesson summaries, quiz questions, transcripts, and translations directly from lecture material. These systems also analyse learner interaction data to identify where students struggle, triggering the generation of supplementary explanations or practice exercises.

The disruptive impact lies in continuous, data-driven content production. Instead of producing fixed content assets periodically, organisations operate automated pipelines where data triggers generation, models produce multiple content variants, and analytics systems optimise performance.

In the generative AI era, content operations increasingly resemble software pipelines – automated, scalable, and capable of producing large volumes of personalised content with minimal manual coordination.

Video 11: Redefining Professional Services

For decades, industries such as law, consulting, and accounting operated on a similar economic model. Expertise was scarce, work was labour-intensive, and services were typically billed by the hour. Clients paid for teams of professionals to research information, analyse documents, and prepare structured advice.

Generative AI is beginning to disrupt this model by embedding knowledge directly into digital systems, reducing the need for traditional information intermediaries and changing how professional expertise is delivered at scale.

In legal services, firms such as Clifford Chance are deploying generative AI systems trained on large repositories of contracts, case law, and regulatory documents. Lawyers can query these systems conversationally, asking the model to summarise key clauses across hundreds of agreements or identify regulatory obligations across multiple jurisdictions.

Behind the scenes, large language models perform semantic search, document summarisation, and clause comparison across large volumes of legal text. Tasks that previously required teams of junior lawyers reviewing documents manually can now be completed far more quickly through AI-assisted analysis.

Importantly, the technology does not replace legal judgement. Senior lawyers still validate interpretations, assess legal risk, and advise clients strategically. However, the time required for routine legal analysis is dramatically reduced.

Consulting firms are integrating generative AI into knowledge workflows in a similar way. At McKinsey & Company, internal AI assistants help consultants synthesise research reports, summarise market intelligence, and generate early drafts of presentations and strategic recommendations.

These systems combine large language models with proprietary knowledge bases, internal documents, and analytics tools. Consultants can therefore analyse complex industry information and generate structured insights significantly faster than through traditional research-heavy consulting workflows.

The economic impact is substantial. Instead of spending days consolidating research findings manually, teams can move more quickly towards interpretation, strategic framing, and client discussion. Human expertise shifts from information gathering towards higher-value advisory and decision support.

Accounting and audit functions are evolving in a similar direction. Firms such as PwC increasingly use AI-driven systems to analyse entire financial datasets rather than relying heavily on manual sampling methods.

Machine learning models scan transaction records, identify anomalies, flag potential compliance risks, and detect unusual financial reporting patterns across large volumes of data. This allows auditors to focus attention on higher-risk areas while improving both speed and analytical coverage.

The broader disruption lies in knowledge disintermediation. Tasks that once required large teams – document review, research synthesis, financial analysis, or information retrieval – can increasingly be performed by AI systems embedded within professional workflows.

As a result, professional services firms are shifting from labour-intensive delivery models towards AI-augmented advisory models. Human professionals focus more on judgement, client relationships, negotiation, and complex interpretation, while AI systems handle large-scale analysis and knowledge retrieval.

In the generative AI era, the value of professional expertise increasingly lies not in simply accessing information, but in interpreting, contextualising, and applying machine-generated insight strategically for clients and organisations.

Video 12: Platformisation of Expertise

Imagine needing a financial model, a legal clause review, or a brand strategy. Ten years ago, you would likely hire a consulting firm or agency and wait weeks for a team to deliver the work. Today, the same expertise can often be accessed within hours through digital platforms. This shift reflects a broader transformation known as the platformisation of expertise.

Generative AI is accelerating this transition by enabling experts to scale their capabilities and deliver specialised services through online marketplaces.

Consider Upwork. The platform has integrated AI tools that help freelancers analyse client briefs, generate proposal drafts, and structure deliverables more efficiently. AI-assisted matching systems also recommend experts based on skills, ratings, and past outcomes.

As a result, companies can quickly hire niche specialists – such as a compliance analyst, data modeller, or instructional designer – for narrowly defined tasks rather than engaging full consulting teams.

A similar trend is visible on Fiverr, where professionals increasingly offer AI-augmented micro-services. Designers generate brand assets using generative design tools, marketers produce campaign content rapidly, and analysts automate reporting workflows.

AI effectively multiplies the productivity of individual professionals, enabling a single expert to deliver work that previously required a small team.

The result is a structural change in how expertise is delivered. Complex projects are decomposed into modular problem units, and organisations assemble temporary networks of specialists to solve them.

In this model, expertise becomes searchable, on-demand, and scalable through platforms.

In the generative AI era, competitive advantage will not depend only on employing large teams of experts. It will increasingly depend on orchestrating AI-augmented expert ecosystems through digital platforms.

Video 13: Final Thoughts

Across this week, we have examined Generative AI through a disruption lens – understanding how it extends beyond incremental improvement to reshape how value is created across organisations. We began by exploring what makes an innovation truly transformative, recognising Generative AI as a capability discontinuity and a general-purpose technology with cross-functional and platform-level implications.

We saw the shift from product-centric to intelligence-centric firms, the blurring of industry boundaries, and the emergence of zero-marginal-cost knowledge models. In marketing and customer interfaces, AI enables hyper-personalisation, conversational engagement, and customer co-creation.

In operations, it compresses design cycles, enables simulation through digital twins, and supports scalable, AI-driven content and expertise platforms.

Taken together, these shifts signal a deeper transition – where AI becomes an embedded intelligence layer shaping how organisations operate, compete, and evolve.

In the upcoming week, we will build on this foundation by examining how these disruptions translate into new business models, organisational structures, and strategic challenges, and what it means to operate as an AI-native enterprise.